

Verified Operational Semantics in Logical Form: A Lean 4 Formalization of the OSLF Algorithm (DRAFT --- July 18, 2026)

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July 18, 2026

Abstract

We present a comprehensive Lean 4 formalization of Operational Semantics in Logical Form (OSLF), the algorithm that mechanically derives spatial-behavioral type systems from rewrite rules. Our formalization spans 44,000+ lines across 124 Lean files (95 in the core OSLF tree, 29 in the process-calculi layer), with 0 sorries and 0 custom axioms across the entire pipeline. The formalization covers: (i) MeTTaIL, a meta-language for defining process calculi with capture-avoiding substitution and a totalized pattern matcher; (ii) the ρ -calculus with full reduction semantics, structural congruence, and modal type system; (iii) the abstract OSLF framework instantiated for *seven* languages (ρ -calculus, lambda calculus, Petri nets, TinyML, MeTTaMinimal, MeTTaFull, and PathMap), each with *fully proven* Galois connections $\Diamond \dashv \blacksquare$; (iv) executable rewrite engines (specialized and generic), including premise-aware rewriting with pluggable relation environments, proven sound with respect to declarative reduction specifications; (v) a *constructor category* built from sort-crossing constructors with a `SubobjectFibration` and `ChangeOfBase`, connecting to the GSLT categorical infrastructure; (vi) *derived typing rules* where the modal operator ($\Diamond$ or $\blacksquare$) assigned to each constructor is determined automatically by its position in the constructor category; (vii) a *presheaf-primary categorical lift* with interface-selected base categories, representable-fiber bridges, and graph-object reduction semantics; (viii) a *Beck-Chevalley analysis* of substitution as change-of-base, with a proven counterexample showing the strong condition fails and concrete representable/graph square theorems; (ix) a MeTTa Core interpreter specification with grounded-operation determinism and state-machine invariants; and (x) *paper-parity endpoints* providing a unified canonical package (`coreMain_paper_parity_full_package`) that bundles Theorem 1 forward substitutability, full presheaf Grothendieck category, scoped/full comparison, TOGL graph-modal bridge, and internal-language bridge for any language instance. All Galois connections, the type soundness theorem, the engine soundness theorem, the constructor fibration, the derived typing rules, the Beck-Chevalley analysis, and the π -to- ρ encoding all carry zero sorries.

1 Introduction

The OSLF algorithm [?] takes a rewrite system as input and produces a spatial-behavioral type system as output. The core insight is that every reduction relation induces a pair of adjoint modal operators:

$$\Diamond\varphi = \{p \mid \exists q. p \rightsquigarrow q \wedge q \in \varphi\} \quad (\text{step-future / possibly}) \quad (1)$$

$$\blacksquare\varphi = \{q \mid \forall p. p \rightsquigarrow q \Rightarrow p \in \varphi\} \quad (\text{step-past / rely}) \quad (2)$$

and that $\Diamond \dashv \blacksquare$ forms a Galois connection. Combined with the spatial decomposition from parallel composition, this yields a type system where types are “behavioral neighborhoods” and typing is substitutability under bisimulation.

Previous treatments of OSLF were paper-only. We give the first machine-checked formalization, connecting:

- a *generic* abstract framework (any rewrite system),
- a *concrete* ϱ -calculus instance with all rules proven,
- a *categorical* derivation via a constructor category with fibered change-of-base and derived typing rules,
- an *executable* reduction engine proven sound w.r.t. the spec, and
- *six language instances* validating the full pipeline.

Contributions.

1. A Lean 4 formalization of the OSLF algorithm as an abstract structure (`RewriteSystem` $\rightarrow$ `OSLFTypeSystem`) and its full ϱ -calculus instance (`rhoOSLF`) with a proven Galois connection.
2. A categorical proof that the Galois connection arises from the adjoint triple $\exists_f \dashv f^* \dashv \forall_f$ applied to the reduction span, with the result shown equal to the direct ϱ -calculus modalities.
3. A *constructor category* built from sort-crossing constructors of any `LanguageDef`, with a `SubobjectFibration` and `ChangeOfBase` connecting to the GSLT infrastructure.
4. *Derived typing rules*: the modal operator ($\Diamond/\blacksquare/\text{id}$) assigned to each constructor is determined automatically by its classification (quoting/reflecting/neutral), and the assignment is proven correct for the ϱ -calculus.
5. A *Beck–Chevalley analysis* of the COMM rule as change-of-base along the substitution map, with a proven counterexample showing the GSLT strong Beck–Chevalley condition does not hold for the constructor fibration, plus representable-fiber and graph-object square theorems consumed by checker-facing corollaries.

6. Executable reduction engines handling the paper-faithful COMM core and context descent, with machine-checked soundness theorems.
7. A least authored contextual reduction relation (`ContextualStep.Step`) independent of its fuel-indexed compiler, with exact soundness and completeness and no representation-wide descent policy.
8. *Six* OSLF instantiations validating generality: ϱ -calculus, lambda calculus, Petri nets, and TinyML (a multi-sort CBV λ -calculus with booleans, pairs, and thunks), plus MeTTaMinimal and MeTTaFull state-machine clients.
9. MeTTaIL: a meta-language for defining process calculi, with totalized pattern matching, 29 proven theorems about capture-avoiding substitution, and a complete ϱ -calculus language definition.
10. A verified bounded model checker for OSLF formulas, with support for predecessor-based $\blacksquare$ checking and proven soundness.
11. A MeTTa Core interpreter specification with grounded-operation determinism, knowledge-preservation invariants, and state bisimilarity (0 sorries; see caveat in Section 7.2).
12. A dedicated sorry-free core entrypoint (`CoreMain.lean`) and machine-readable FULL tracker (`Framework/FULLStatus.lean`) for review.
13. *A paper-parity canonical package* (`coreMain_paper_parity_full_package`) unifying Theorem 1 (substitutability), the full presheaf Grothendieck category, scoped/full comparison, TOGL graph-modal bridge, and internal-language bridge into a single endpoint.
14. *A strict paper-claim tracker* (`Framework/PaperClaimTracker.lean`) mapping 27 verifiable claims from three source papers to concrete Lean theorem names, with 25 fully proven and 2 assumption-scoped. A separate *strict NTT claim tracker* (`Framework/NTTClaimTracker.lean`) keys 12 claims to Native Type Theory theorem numbers; all 12 are resolved (11 proven, 1 assumption-scoped with proven necessity).
15. *Assumption-necessity counterexamples* (`Framework/AssumptionNecessity.lean`) proving that the `types.Nonempty` guard on Π/Σ formation and the `hClosed` fragment-closure hypothesis cannot be discharged in general.
16. *Robustness canaries* (`Framework/PaperParityCanaries.lean`) instantiating the canonical package on ϱ -calculus and λ -calculus, plus a negative canary showing `hClosed` does real work.

Scope Boundary and Non-Claims. This draft reports what is machine-checked in the current Lean corpus; it is not a claim that every construction discussed across the broader literature folders has been fully formalized end-to-end. In particular, the cited OSLF and Native Type Theory papers explicitly list larger future directions that remain outside the current core claim surface, including:

- full internal natural-transformation presentation of modalities ($\Omega \rightarrow \Omega$ endomorphisms) over the presheaf route,
- full premise-rich Lean $\rightarrow$ Rust ingestion for MeTTa exports; current roundtrip validation covers TinyML smoke and the premise-free MeTTaMinimal export slice,
- larger extensions such as stochastic and quantum variants listed as future-work directions in the source papers.

Note: the Grothendieck construction for Native Type Theory, the internal language (frame-derived $\rightarrow/\neg, \top/\perp/\wedge/\vee$), and the TOGL graph-modal bridge are now formalized as theorem-level endpoints bundled in `coreMain_paper_parity_full_package`; the Π/Σ type formation is assumption-scoped (`types.Nonempty` guard, proven necessary). Accordingly, this document distinguishes *verified core endpoints* from *research-frontier extensions*, and the strict paper-alignment status is tracked explicitly in `Mettapedia/OSLF/Framework/FULLStatus.lean`.

2 Background: The ϱ -Calculus

The reflective higher-order calculus [?] extends the π -calculus with:

- *Quoting*: any process P can be turned into a name $@P$ (name = quoted process).
- *Dequoting*: $*x$ recovers the process quoted by name x .
- *No built-in names*: all names arise from quoting, giving the calculus a reflective character.

The reduction rules are:

$$\text{COMM: } \{n!(q) \mid \mathbf{for}(x \leftarrow n)\{p\} \mid \mathit{rest}\} \rightsquigarrow \{p[@q/x] \mid \mathit{rest}\} \quad (3)$$

$$(4)$$

plus structural congruence (11 rules) and contextual reduction under parallel composition. Free drop is inert in the core relation and only becomes executable through semantic substitution when a quoted process is supplied to a bound name.

3 Formalization Architecture

3.1 Module Structure

The formalization is organized in seven directories plus standalone files:

Component	Lines	Sorries	Content
MeTTaIL/	3,592	0	Meta-language AST, substitution, matching, declarative
MeTTaCore/	3,839	0	Interpreter specification
Framework/	14,860	0	Abstract OSLF + categorical bridge + 6 instances
PathMap/	1,890	0	Quantale interface + <code>Finset</code> instance + complete NTT
OSLF/RhoCalculus/	60	0	OSLF ϱ -calculus facades
NativeType/	1,775	0	Native type construction + Grothendieck + strengthened
Formula.lean	1,204	0	Verified bounded model checker
Main/CoreMain/SpecIndex	2,001	0	Focused OSLF re-exports + spec index
Languages/PiCalculus/	8,768	0	π -calculus + ϱ -encoding
Languages/RhoCalculus/	4,848	0	Concrete ϱ -calculus + engine
Total	42,249	0	0 sorries, 0 axioms

Operational entrypoints (`CoreMain.lean` and `Main.lean`) remain on the core boundary, with process-calculus facades exposed under `Mettapedia/Languages/ProcessCalculi*.lean`. The $\pi \rightarrow \varrho$ encoding (forward simulation, backward normalization, weak bisimilarity, and encoding morphism) is fully proven with 0 sorries; for detailed status see `Airxiv/airxiv_2026-02-19_pi_rho_embedding_status.tex`.

3.2 Canonical Endpoint Map

For review and downstream reuse, the canonical API surface is concentrated in `Mettapedia/OSLF/CoreMain.lean` and `Mettapedia/OSLF/Framework/PiRhoCanonicalBridge.lean`. The exact endpoint names are:

CoreMain-facing entrypoints

- `piRho_coreMain_canonical_contract_end_to_end`
- `coreMain_piRho_canonical_contract`
- `coreMain_piRho_contract_projection_api`
- `coreMain_nativeType_piOmega_translation_endpoint`
- `coreMain_nativeType_piOmegaProp_translation_endpoint`
- `coreMain_nativeType_piProp_colax_rules_endpoint`
- `coreMain_nativeType_full_presheaf_morphism_endpoint`
- `coreMain_nativeType_scoped_full_constructor_comparison_package`

- `coreMain_category_topos_package`
- `coreMain_topos_internal_language_bridge_package`
- `coreMain_togl_graph_modal_bridge_package`
- `coreMain_section12_worked_examples`
- `coreMain_dependent_parametric_generated_typing`
- `coreMain_paper_parity_full_package`
- `coreMain_paper_parity_canonical_package`

Bridge-level canonical contracts

- `PiRhoCoreMainCanonicalContract`
- `piRho_coreMain_predDomain_endpoint`
- `piRho_coreMain_predDomain_transfer_bundle_end_to_end`
- `piRho_coreMain_predDomain_reachable_state_end_to_end`
- `reachableCoreStarFiniteSubrelation`
- `reachableDerivedStarFiniteSubrelation`
- `hm_converse_rhoCoreCanonicalRel`
- `hm_converse_rhoDerivedCanonicalRel`
- `ToposTOGLBridge.topos_internal_language_bridge_package`
- `ToposTOGLBridge.togl_graph_modal_bridge_package`
- `ToposTOGLBridge.graphChainN_iff_relCompN`
- `ToposTOGLBridge.diamondIterN_iff_graphChainN`
- `ToposTOGLBridge.togl_internal_graph_correspondence_layer`
- `NativeType.fullPresheafGrothendieckCategory`
- `NativeType.scoped_full_scoped_obj_roundtrip`
- `AssumptionNecessity.types_nonempty_necessary_for_piSigma`
- `AssumptionNecessity.hClosed_necessary_for_fragment`
- `AssumptionNecessity.not_global_hImageFinite_rhoCoreStarRel`
- `NativeType.prop12_package` (indexed adjoints + Beck–Chevalley)
- `NativeType.prop14_cosmicFibration` (Frame fibers)

- `NativeType.prop17_reification` (reification right adjoint)
- `NativeType.def21_cartesianLift_proj` (Cartesian lift projection)
- `NativeType.def21_cartesianLift_universal_comp` (universal factorization)
- `NativeType.imageComprehension_iff` (hom-set characterization $i \dashv c$)
- `NativeType.thm23_functorialLaws` (identity/composition laws)
- `NTTClaimTracker.nttRemainingCount_zero` (strict NTT parity closed)

The `Framework/` directory (14,708 lines, 32 files) is the largest component.
Key files:

File	Lines	Content
<code>PiRhoCanonicalBridge.lean</code>	1,876	$\pi \rightarrow \varrho$ canonical bridge
<code>CategoryBridge.lean</code>	1,603	Presheaf topos, Yoneda, Ω classifier
<code>BeckChevalleyOSLF.lean</code>	1,233	Substitution as change-of-base
<code>TinyMLInstance.lean</code>	758	CBV λ -calculus with booleans/pairs/thunks
<code>PLNSelectorLanguageDef.lean</code>	723	PLN selector as <code>LanguageDef</code>
<code>MeTTaMinimalInstance.lean</code>	700	MeTTa state-machine client
<code>ConstructorCategory.lean</code>	460	Sort quiver + free category
<code>TypeSynthesis.lean</code>	441	<code>langOSLF</code> pipeline
<code>PLNSelectorGSLT.lean</code>	373	PLN selector rules as OSLF/GSLT
<code>DerivedTyping.lean</code>	346	Generic typing rules from constructor category
<code>MeTTaFullInstance.lean</code>	346	Full MeTTa language definition
<code>ModalEquivalence.lean</code>	353	Constructor change-of-base $\leftrightarrow$ modalities
<code>GeneratedTyping.lean</code>	474	<code>GenHasType</code> typing rules
<code>SynthesisBridge.lean</code>	294	Three-layer bridge
<code>LanguageMorphism.lean</code>	506	Language translation infrastructure
<code>ConstructorFibration.lean</code>	251	<code>SubobjectFibration</code> + <code>ChangeOfBase</code>
<code>DerivedModalities.lean</code>	250	Adjoint triple derivation
<code>PetriNetInstance.lean</code>	233	Petri net OSLF instance
<code>LambdaInstance.lean</code>	218	Lambda calculus OSLF instance
<code>RewriteSystem.lean</code>	196	Abstract OSLF input/output
<code>RhoInstance.lean</code>	134	ϱ -calculus instance
<code>ToposTOGLBridge.lean</code>	767	TOGL graph-modal + internal language
<code>ToposReduction.lean</code>	401	Graph-object reduction semantics
<code>FULLStatus.lean</code>	554	Machine-readable paper-parity tracker
<code>AssumptionNecessity.lean</code>	279	Assumption-necessity counterexamples
<code>PaperClaimTracker.lean</code>	156	Strict 27-claim paper-claim tracker
<code>NTTClaimTracker.lean</code>	128	Strict 12-claim NTT-keyed tracker
<code>PaperParityCanaries.lean</code>	104	Robustness canaries (positive + negative)

3.3 Abstract OSLF Framework

The abstract layer defines two key structures:

Listing 1: The OSLF input and output (RewriteSystem.lean)

```
structure RewriteSystem where
  Sorts      : Type*
  procSort   : Sorts
  Term       : Sorts -> Type*
  Reduces    : Term procSort -> Term procSort -> Prop

structure OSLFTypeSystem where
  Sorts      : Type*
  procSort   : Sorts
  Term       : Sorts -> Type*
  Pred       : Sorts -> Type*
  [frame     : (S : Sorts) -> Frame (Pred S)]
  satisfies  : (S : Sorts) -> Term S -> Pred S -> Prop
  diamond    : Pred procSort -> Pred procSort
  box        : Pred procSort -> Pred procSort
  galois     : GaloisConnection diamond box
```

3.4 The Galois Connection

The central theorem: $\Diamond \dashv \blacksquare$.

Theorem 1 (Galois Connection, 0 sorries). *For all predicates φ, ψ on $\mathcal{Q}$ -calculus processes:*

$$\Diamond \varphi \leq \psi \iff \varphi \leq \blacksquare \psi$$

where $\Diamond \varphi(p) = \exists q. p \rightsquigarrow q \wedge \varphi(q)$ and $\blacksquare \psi(q) = \forall p. p \rightsquigarrow q \rightarrow \psi(p)$.

In Lean:

Listing 2: The Galois connection (Reduction.lean)

```
theorem galois_connection :
  GaloisConnection possiblyProp relyProp := by
  intro phi psi
  constructor
  . intro h q hrely p hred
    exact h (hrely p hred)
  . intro h p hposs
    exact h.2 p hposs.1 hposs.2
```

3.5 Categorical Derivation via Adjoint Triples

The Galois connection arises from the general theory of change-of-base along a span.

Definition 2 (Reduction Span). *A span $\mathcal{S} \xleftarrow{\text{src}} E \xrightarrow{\text{tgt}} \mathcal{S}$ where E is the set of reduction edges, src extracts the source, and tgt extracts the target.*

From any such span we derive three operations on predicates:

$$f^*(\psi)(e) = \psi(\text{tgt}(e)) \quad (\text{pullback}) \quad (5)$$

$$\exists_f(\varphi)(q) = \exists e. \text{tgt}(e) = q \wedge \varphi(e) \quad (\text{direct image}) \quad (6)$$

$$\forall_f(\varphi)(q) = \forall e. \text{tgt}(e) = q \rightarrow \varphi(e) \quad (\text{universal image}) \quad (7)$$

Theorem 3 (Derived Galois, 0 sorries). *For any **ReductionSpan**, the composition $\diamond = \exists_{\text{src}} \circ \text{tgt}^*$ and $\blacksquare = \forall_{\text{tgt}} \circ \text{src}^*$ form a Galois connection. Furthermore, for the ρ -calculus span, the derived operators equal the concrete **possiblyProp** and **relyProp**.*

Listing 3: Derived modalities equal concrete (DerivedModalities.lean)

```
theorem derived_diamond_eq_possiblyProp :
  derivedDiamond rhoSpan = possiblyProp := ...

theorem derived_box_eq_relyProp :
  derivedBox rhoSpan = relyProp := ...

theorem rho_galois_from_span :
  GaloisConnection (derivedDiamond rhoSpan)
    (derivedBox rhoSpan) :=
  derived_galois rhoSpan
```

4 Constructor Category and Fibration

A key contribution of this formalization is the *constructor category*: a non-discrete category built from the sort-crossing constructors of any **LanguageDef**, replacing the discrete **Discrete R.Sorts** from the earlier categorical lift.

4.1 Sort Quiver and Free Category

Given a **LanguageDef**, we extract the *unary sort-crossing constructors*: grammar rules with exactly one **.simple** parameter whose base sort differs from the constructor's output sort. These become the arrows of a quiver on the language's sorts.

Listing 4: Constructor category (ConstructorCategory.lean)

```
-- Sort type: valid sort names
def LangSort (lang : LanguageDef) :=
  { s : String // s IN lang.types }

-- Sort-crossing arrows
structure SortArrow (lang : LanguageDef)
  (dom cod : LangSort lang) where
  label : String
  valid : (label, dom.val, cod.val) IN unaryCrossings lang

-- Free category: paths of sort-crossing arrows
```

```

inductive SortPath (lang : LanguageDef)
  : LangSort lang -> LangSort lang -> Type where
| nil   : SortPath lang s s
| cons  : SortPath lang s t -> SortArrow lang t u
        -> SortPath lang s u

```

For the $\mathcal{Q}$ -calculus: 2 objects (Proc, Name), 2 arrows (NQuote: Proc $\rightarrow$ Name, PDrop: Name $\rightarrow$ Proc), and composites PDrop $\circ$ NQuote and NQuote $\circ$ PDrop.

Each arrow has a *semantic function* `arrowSem`: wrapping a pattern in the constructor's `.apply` node (e.g., $p \mapsto \text{NQuote}(p)$). This extends to paths via `pathSem`, with a proven composition law `pathSem_comp`.

A *universal property* (free category lifting) is proven: any assignment of objects and arrows to a target category $\mathcal{C}$ lifts uniquely to a functor `liftFunctor`, with uniqueness proven in `lift_map_unique`.

4.2 SubobjectFibration and ChangeOfBase

Over the constructor category we build a fibration and change-of-base (`Framework/ConstructorFibration.lean`, 251 lines, 0 sorries):

Listing 5: Constructor fibration (`ConstructorFibration.lean`)

```

-- Each sort has fiber Pattern -> Prop (a Frame)
def constructorFibration (lang : LanguageDef) :
  SubobjectFibration (ConstructorObj lang) where
  Sub := fun _ => Pattern -> Prop
  frame := fun _ => Pi.instFrame

-- Full change-of-base with proven adjunctions
def constructorChangeOfBase (lang : LanguageDef) :
  ChangeOfBase (constructorFibration lang) where
  pullback f      := pb (pathSem lang f)
  directImage f   := di (pathSem lang f)
  universalImage f := ui (pathSem lang f)
  direct_pullback_adj f := di_pb_adj (pathSem lang f)
  pullback_universal_adj f := pb_ui_adj (pathSem lang f)
  ...

```

The adjunctions $\exists_f \dashv f^* \dashv \forall_f$ are proven (not axiomatized), following from the generic `di_pb_adj` / `pb_ui_adj` in `DerivedModalities.lean`.

Key proven properties:

- **Pullback functoriality:** $(f \circ g)^* = g^* \circ f^*$ (from `pathSem_comp`), $id^*(\varphi) = \varphi$.
- **Frame morphism:** $f^*(\varphi \wedge \psi) = f^*(\varphi) \wedge f^*(\psi)$ and $f^*(\top) = \top$ (both by `rfl`).
- **Monotonicity** of all three operations (from adjunctions).

4.3 Modal Equivalence

The file `Framework/ModalEquivalence.lean` (353 lines) connects the constructor change-of-base to the OSLF modalities:

Theorem 4 (Modal = Change-of-Base, 0 sorries). *The OSLF modalities are Set-level change-of-base along the reduction span:*

$$\begin{aligned}\Diamond_{lang} &= \exists_{src} \circ \text{tgt}^* && (\text{definitional}) \\ \blacksquare_{lang} &= \forall_{tgt} \circ \text{src}^* && (\text{definitional})\end{aligned}$$

For the ϱ -calculus, this gives the *typing actions*:

- **NQuote** ($\text{Proc} \rightarrow \text{Name}$): $\varphi \mapsto \Diamond\varphi$ (“can reduce to φ ”)
- **PDrop** ($\text{Name} \rightarrow \text{Proc}$): $\alpha \mapsto \blacksquare\alpha$ (“all predecessors satisfy α ”)

The composite $\text{PDrop} \circ \text{NQuote}$ gives $\blacksquare \circ \Diamond$ and $\text{NQuote} \circ \text{PDrop}$ gives $\Diamond \circ \blacksquare$. The typing action Galois connection $\Diamond \dashv \blacksquare$ is proven as an instance of the general language Galois connection.

4.4 Derived Typing Rules

The file `Framework/DerivedTyping.lean` (346 lines, 0 sorries) derives typing rules generically from the constructor category structure.

Each sort-crossing arrow is automatically classified:

Listing 6: Constructor classification (`DerivedTyping.lean`)

```
inductive ConstructorRole where
| quoting    -- domain = procSort: introduces diamond
| reflecting -- codomain = procSort: introduces box
| neutral    -- neither: identity

def classifyArrow (lang : LanguageDef) (procSort : String)
  (arr : SortArrow lang dom cod) : ConstructorRole :=
  if dom.val = procSort then .quoting
  else if cod.val = procSort then .reflecting
  else .neutral
```

Theorem 5 (Classification Correctness, 0 sorries). *For the ϱ -calculus:*

- *NQuote is classified as quoting; its typing action equals $\Diamond$.*
- *PDrop is classified as reflecting; its typing action equals $\blacksquare$.*

The `DerivedHasType` judgment provides a generic typing rule for unary sort-crossing constructors: apply the constructor’s typing action ($\Diamond/\blacksquare/\text{id}$) to the argument’s predicate, then tag the result at the output sort.

4.5 Beck–Chevalley for Substitution

The file `Framework/BeckChevalleyOSLF.lean` (1,233 lines, 0 sorries) analyzes the COMM rule’s substitution $p[@q/x]$ as a change-of-base map.

Definition 6 (COMM Substitution Map). $\sigma_q : \text{Pattern} \rightarrow \text{Pattern}$ defined by $\sigma_q(pBody) = \text{openBVar } 0 \ (\text{NQuote}(q)) \ pBody$.

This induces the adjoint triple $\exists_{\sigma_q} \dashv \sigma_q^* \dashv \forall_{\sigma_q}$ via the same `pb/di/ui` infrastructure, and the modal+substitution Galois connections compose:

Theorem 7 (Composed Galois, 0 sorries). $\Diamond \circ \exists_{\sigma_q} \dashv \sigma_q^* \circ \blacksquare$

The COMM rule’s type preservation (`comm_preserves_type` from `Soundness.lean`) is re-expressed categorically as a pullback inequality:

Theorem 8 (Substitutability as Pullback, 0 sorries). *For any typing context Γ , type τ , variable x , value q , and type σ with $\Gamma \vdash q : \sigma$:*

$$\text{typedAt}(\Gamma[x \mapsto \sigma], \tau) \leq \sigma_q^*(\text{typedAt}(\Gamma, \tau))$$

The COMM substitution map factors through the constructor semantics: $\sigma_q(p) = \text{openBVar } 0 \ (\text{pathSem } \text{nquoteMor } q) \ p$.

Theorem 9 (Strong Beck–Chevalley Fails, 0 sorries). *The GSLT’s universal Beck–Chevalley condition $f^* \circ \exists_g = \exists_{\pi_1} \circ \pi_2^*$ does **not** hold for the constructor fibration.*

Concretely, for the commuting square $\text{PDrop} \circ \text{NQuote} = \text{PDrop} \circ \text{NQuote}$:

$$\text{PDrop}^*(\exists_{\text{PDrop}}(\top))(fvar\ x) = \top$$

but

$$\exists_{\text{NQuote}}(\text{NQuote}^*(\top))(fvar\ x) = \perp$$

because $\text{NQuote}(q) \neq fvar\ x$ for all q .

The counterexample is proven by exhibiting a concrete witness at `fvar “x”`: the LHS is inhabited by $\langle fvar\ x, rfl, \top \rangle$ while the RHS requires some p with $\text{NQuote}(p) = fvar\ x$, which is impossible since $\text{NQuote}(p) = \text{.apply “NQuote” } [p]$.

5 Executable Rewrite Engine

A formalization that only *specifies* reduction is incomplete for verification of actual implementations. We provide `reduceStep`, a computable function that enumerates all one-step reducts, proven sound w.r.t. the propositional `Reduces`.

5.1 Engine Design

Listing 7: The executable engine (Engine.lean)

```
def reduceStep (p : Pattern) (fuel : Nat := 100)
  : List Pattern :=
  match fuel with
  | 0 => []
  | fuel + 1 =>
    match p with
    | .collection .hashBag elems none =>
      let commReducts := findAllComm elems
      let parReducts :=
        reduceElemsAux (reduceStep . fuel) elems
        |>.map fun (i, elem') =>
          .collection .hashBag (elems.set i elem') none
      commReducts ++ parReducts
    | .apply "PDrop" [.apply "NQuote" [inner]] =>
      [inner]
    | _ => []
```

The engine handles COMM (all output-input pairs on matching channels), DROP ($*(@P) \rightsquigarrow P$), and PAR (recursive reduction under parallel composition). Non-deterministic races produce multiple reducts in the output list.

5.2 Soundness

Theorem 10 (Engine Soundness, 0 sorries). *Every reduct computed by `reduceStep` corresponds to a valid `Reduces`:*

$$q \in \text{reduceStep}(p, \text{fuel}) \implies \exists d : p \rightsquigarrow q$$

The proof proceeds by case analysis:

- **COMM:** Each output-input pair is located by `findAllComm`, whose specification is proven to identify valid COMM redex positions. The soundness chain is: list permutation (`perm_extract_two`) $\rightarrow$ structural congruence (`par_perm`) $\rightarrow$ `Reduces.comm` $\rightarrow$ `Reduces.equiv`.
- **PAR:** The `reduceElemsAux_spec` lemma extracts the sub-element index and recursive reduct. List decomposition via `List.set_eq_take_append_cons_drop` connects `List.set` to the *before*++*[p]*++*after* form required by `Reduces.par_any`.

6 Generic MeTTaIL Rewrite Framework

The specialized ϱ -calculus engine is lifted to a language-parametric framework. Given any `LanguageDef` (a list of sorts, constructors, equations, and rewrite rules), the generic engine automatically:

1. matches concrete terms against rule LHS patterns (multiset matching with rest variables),

2. produces variable bindings via alpha-renaming for binders,
3. applies bindings to rule RHS patterns (including substitution evaluation),
and
4. iterates under congruence (subterm rewriting inside parallel compositions).

Listing 8: Premise-aware generic rewrite step (Engine.lean)

```

structure RelationEnv where
  tuples : String -> List Pattern -> List (List Pattern)

def applyRuleWithPremisesUsing
  (relEnv : RelationEnv) (lang : LanguageDef)
  (rule : RewriteRule) (term : Pattern) : List Pattern :=
  (matchPattern rule.left term).flatMap fun bs =>
    (applyPremisesWithEnv relEnv lang rule.premises bs).map fun bs' =>
      applyBindings bs' rule.right

def rewriteStepWithPremisesUsing
  (relEnv : RelationEnv) (lang : LanguageDef) (term : Pattern) : List
  Pattern :=
  lang.rewrites.flatMap fun rule => applyRuleWithPremisesUsing relEnv
  lang rule term

```

Multiset matching. The key algorithmic contribution is `matchBag`: for a collection pattern with n elements and an optional rest variable, it enumerates all ways to match the n pattern elements against term elements (backtracking search over permutations), binding unmatched elements to the rest variable.

Authored contextual reduction. The file `MeTTaIL/ContextualStep.lean` defines the independently inductive relation `ContextualStep.StepAt` and its unbounded least relation `ContextualStep.Step`. Contextual authority is carried only by authored `Premise.congruence` premises. The fuel-indexed compiler `rewriteAt` is proven both *sound* and *complete* at every depth, and existentially exact for the unbounded relation (0 sorries).

7 MeTTaIL and MeTTa Core

7.1 MeTTaIL: The Meta-Language

MeTTaIL (“MeTTa Intermediate Language”) defines the AST shared by all process calculi in the formalization, using locally nameless representation. The core type is `Pattern` with 7 constructors:

Listing 9: Pattern AST — locally nameless (Syntax.lean)

```

inductive Pattern where
| bvar      : Nat -> Pattern      -- bound variable (de Bruijn)
| fvar      : String -> Pattern  -- free variable / metavariable

```

```

| apply      : String -> List Pattern -> Pattern
| lambda     : Pattern -> Pattern      -- binder (no name)
| multiLambda: Nat -> Pattern -> Pattern
| subst      : Pattern -> Pattern -> Pattern
| collection : CollType -> List Pattern
              -> Option String -> Pattern

```

Key proven properties of substitution (29 theorems, 0 sorries):

- `subst_empty`: empty substitution is the identity;
- `subst_fresh`: substitution on a fresh variable is the identity;
- `commSubst`: the COMM-rule substitution $p[@q/x]$ respects freshness.

7.2 MeTTa Core Interpreter

The `MeTTaCore/` directory provides a complete interpreter specification for Hyperon Experimental MeTTa (4,419 lines, 0 sorries), including:

- `Atom`: the universal term type with `DecidableEq`;
- `Bindings`: variable resolution with merge and transitive lookup;
- `MeTTaState`: the 4-register $\langle i, k, w, o \rangle$ machine;
- `PatternMatch`: bidirectional unification;
- `MinimalOps`: grounded operations ($+$, $-$, $*$, $/$, $<$, etc.);
- `RewriteRules`: equation-driven rewriting;
- `Atomspace`: multiset-based knowledge base with query operations;
- `Properties`: grounded-operation determinism, knowledge-preservation invariants, and state bisimilarity (see below).

Properties caveat. The `Properties.lean` theorems are weaker than their names suggest:

- `progress`: every workspace term is either insensitive or has matching equations—but this follows directly from the definition of `insensitive` (which checks `queryEquations =  $\emptyset$`), so it is not progress in the type-theoretic sense.
- `grounded_confluence`: determinism of `executeGroundedOp` (same inputs $\Rightarrow$ same output), not confluence of the rewrite system.
- `Bisimilar`: structural equality of output, knowledge, and workspace fields—not barbed bisimulation in the process-calculus sense.

Full rewrite-system confluence and genuine barbed bisimulation remain open.

8 Type Soundness

Theorem 11 (Substitutability, 0 sorries). *If P and Q are bisimilar processes, then they have the same native types:*

$$P \sim Q \implies \forall \tau. P : \tau \iff Q : \tau$$

Theorem 12 (Type Preservation, 0 sorries). *The COMM and DROP rules preserve types:*

$$\Gamma \vdash P : \tau \wedge P \rightsquigarrow Q \implies \Gamma \vdash Q : \tau$$

Both theorems are proven in `RhoCalculus/Soundness.lean` with a 10-constructor typing judgment `HasType` and explicit typing contexts.

9 ϱ -Calculus Instance

The ϱ -calculus is formalized with the paper-faithful core reduction semantics (COMM with structural congruence/PAR closure), structural congruence (11 rules), and multi-step reduction. The OSLF algorithm derives its spatial-behavioral type system, with the Galois connection $\Diamond \dashv \blacksquare$ proven in `RhoCalculus/Soundness.lean` (0 sorries).

10 Type Synthesis: `LanguageDef` $\rightarrow$ `OSLFTypeSystem`

The culmination of the formalization is the *type synthesis pipeline*: given any `LanguageDef`, mechanically produce a full `OSLFTypeSystem` with a **proven Galois connection**.

10.1 The Pipeline

The pipeline proceeds in five steps, all implemented in `Framework/TypeSynthesis.lean`:

1. `langReduces lang p q` wraps the executable engine: $q \in \text{rewriteWithContext } \text{lang } p$.
2. `langRewriteSystem lang` assembles a `RewriteSystem`.
3. `langSpan lang` builds the reduction span (edges = reductions).
4. `langDiamond/langBox` derive modal operators via `derivedDiamond/derivedBox` from the adjoint triple.
5. `langOSLF lang` packages everything into an `OSLFTypeSystem`, with the Galois connection proven by `derived_galois`.

Theorem 13 (Automatic Galois Connection, 0 sorries). *For any `LanguageDef lang`:*

$$\Diamond_{\text{lang}} \dashv \blacksquare_{\text{lang}}$$

where $\Diamond_{\text{lang}} = \exists_{\text{src}} \circ \text{tgt}^*$ and $\blacksquare_{\text{lang}} = \forall_{\text{tgt}} \circ \text{src}^*$ are derived from the reduction span. No manual proof is needed per language.

10.2 Predecessor-Finiteness: Sufficient Condition

The Galois connection $\Diamond \dashv \blacksquare$ holds for every `LanguageDef`, but the full Theorem 1 substitutability equivalence additionally requires predecessor-finiteness of the reduction relation. `Framework/PredFiniteSufficient.lean` (0 sorries) identifies a decidable structural sufficient condition:

Definition 14 (`LangPredFiniteSafe`). *A `LanguageDef` is predecessor-finite-safe when:*

1. *every rewrite rule is variable-preserving (every free variable in the LHS appears in the RHS),*
2. *no rule carries premises,*
3. *both LHS and RHS are `isMatchCorrect` (no `.subst` or `.collection` nodes).*
4. *no reflective presentation changes syntactic matching or substitution.*

Theorem 15 (`predFinite_langReduces_of_langPredFiniteSafe`, 0 sorries). *If `LangPredFiniteSafe lang` then for every term q , the predecessor set $\{p \mid p \rightsquigarrow q\}$ is finite.*

The proof inverts `matchPattern` on the RHS to obtain a finite list of candidate binding sets; variable-preservation ensures all LHS bindings are determined; `applyBindings` on the LHS then reconstructs the unique predecessor. Three structural induction lemmas underpin the argument: completeness, injectivity, and dependence of `applyBindings` with respect to free variables.

Corollary 16 (`theorem1_substitutability_predFiniteSafe`, 0 sorries). *For any `LangPredFiniteSafe` language, Theorem 1 substitutability holds without assuming finiteness as a hypothesis.*

The ρ -calculus specifically fails this condition: `rhoCalc_not_langPredFiniteSafe` (proved by `decide`) shows that the COMM rule's RHS contains `.subst` and `.collection` nodes and erases the channel name n . This complements the negative result `not_global_hPredFinite_langReduces_rhoCalc`.

11 Language Instances

The pipeline is validated by seven language instances of increasing complexity:

11.1 Lambda Calculus (1 sort, 0 crossings)

Untyped lambda calculus with β -reduction (`Framework/LambdaInstance.lean`, 218 lines). One sort (`Term`), two constructors (`App`, `Lam`), one reduction rule. The constructor category is discrete (no sort-crossing constructors). Six executable demos verify β -reduction, multi-step normalization, and formula checking.

11.2 Petri Net (1 sort, 0 crossings)

A simple Petri net with four places and two transitions (`Framework/PetriNetInstance.lean`, 233 lines). Validates that multiset (bag) matching works correctly without any abstraction or substitution machinery. Key properties proven by `native_decide`:

- $\{D\}$ is a dead marking (0 reducts);
- $\{A, B\}$ has exactly 1 reduct via transition T_1 .

11.3 ϱ -Calculus (2 sorts, 2 crossings)

The primary instance with sorts `Proc` and `Name`, constructors `NQuote`: `Proc` $\rightarrow$ `Name` and `PDrop`: `Name` $\rightarrow$ `Proc`. This is the most extensively developed instance, with the full type soundness proof, specialized engine, and Beck-Chevalley analysis (Sections 4–4.5).

11.4 TinyML (2 sorts, 2 crossings, 6 rules)

A call-by-value λ -calculus with booleans, pairs, and thunks (`Framework/TinyMLInstance.lean`, 758 lines, 0 sorries).

Listing 10: TinyML language definition (`TinyMLInstance.lean`)

```
-- Sorts: Expr (process sort), Val (data sort)
-- Constructors:
--   Expr: App, If, Fst, Snd, Inject(v:Val)
--   Val:  BoolT, BoolF, Lam(^body), PairV, Thunk(e:Expr)
-- Reductions:
--   Beta:    App(Inject(Lam(^body)), Inject(v)) ~> body[v/x]
--   Force:   Inject(Thunk(e)) ~> e
--   IfTrue:  If(Inject(BoolT), t, e) ~> t
--   IfFalse: If(Inject(BoolF), t, e) ~> e
--   FstPair: Fst(Inject(PairV(a, b))) ~> Inject(a)
--   SndPair: Snd(Inject(PairV(a, b))) ~> Inject(b)
```

TinyML mirrors the ϱ -calculus sort structure:

TinyML	ϱ-Calculus	Role
<code>Expr</code>	<code>Proc</code>	Process sort (carries reduction)
<code>Val</code>	<code>Name</code>	Data sort
<code>Thunk</code> : <code>Expr</code> $\rightarrow$ <code>Val</code>	<code>NQuote</code> : <code>Proc</code> $\rightarrow$ <code>Name</code>	Quoting ($\diamond$)
<code>Inject</code> : <code>Val</code> $\rightarrow$ <code>Expr</code>	<code>PDrop</code> : <code>Name</code> $\rightarrow$ <code>Proc</code>	Reflecting ($\blacksquare$)
<code>Beta</code>	<code>COMM</code>	Main computation rule
<code>Force</code>	<code>DROP</code>	Quoting/reflecting cancel

The CBV strategy is encoded syntactically: β -reduction requires both the function and argument to be wrapped in `Inject(-)`, ensuring arguments are evaluated to values before substitution.

Key proven results:

- `tinyML_crossings`: exactly 2 sort-crossing constructors (`Inject`, `Thunk`)
– `native_decide`.
- `thunk_is_quoting`: `Thunk` classified as quoting; typing action = $\diamond$.
- `inject_is_reflecting`: `Inject` classified as reflecting; typing action = $\blacksquare$.
- `tinyML_typing_action_galois`: $\diamond \dashv \blacksquare$ for TinyML typing actions.
- 11 executable demos including a 3-step reduction chain (`force` $\rightarrow$ `ifTrue` $\rightarrow$ `fstPair`).

11.5 MeTTaMinimal (2 sorts, 1 crossing, 2 rules)

A minimal MeTTa state-machine client (`Framework/MeTTaMinimalInstance.lean`, 700 lines, 0 sorries). One sort crossing (`Eval`: `Expr` $\rightarrow$ `State`), 2 rules (rewrite and match-fail). Validates the end-to-end path from checker to Beck-Chevalley graph, including `mettaMinimal_checker_to_BC_graph`: the checker's verdicts are compatible with the categorical semantics.

11.6 MeTTaFull (3 sorts, 2 crossings, 4 rules)

A full MeTTa language definition based on the interpreter specification in `MeTTaCore/` (`Framework/MeTTaFullInstance.lean`, 346 lines, 0 sorries). Three sorts (`Atom`, `State`, `Result`), sort-crossing constructors `Interpret`: `Atom` $\rightarrow$ `State` and `Output`: `State` $\rightarrow$ `Result`. Four rewrite rules model the interpret-match-rewrite-output cycle. The end-to-end theorem `mettaFull_checker_to_BC_graph` verifies categorical agreement as with `MeTTaMinimal`.

11.7 PathMap (1 sort, 0 crossings, condition-gated rules)

`PathMap` is an algebraic interface for associative path data structures, originating in the `Hyperon/MeTTa` runtime [?]. The formalization (`PathMap/`, 4,086 lines, 0 sorries) is the first OSLF instance where (a) rewrite rules carry *premise guards* tested against a pluggable `RelationEnv`, and (b) the NTT diamond theorem is a *complete biconditional* rather than a one-sided soundness statement.

Algebraic structure. The core type is

```
inductive AlgebraicResult (V : Type*) where
| none -- empty / absorbing
| identity (self_ident counter_ident : Bool) -- structural identity
| element (v : V) -- genuinely new value
```

Four operations on a value type `V` return `AlgebraicResult V`:

Operation	Intuition	Rust trait
<code>pjoin a b</code>	union (paths in either)	<code>Lattice::pjoin</code>
<code>pmeet a b</code>	intersection (paths in both)	<code>Lattice::pmeet</code>
<code>psubtract a b</code>	difference ($a \setminus b$)	<code>DistributiveLattice</code>
<code>prestrict a b</code>	prefix-filter of a by b	<code>Quantale</code>

Typeclasses `PathMapLattice`, `PathMapDistributiveLattice`, and `PathMapQuantale` axiomatise commutativity, idempotency, absorption, and the `SubtractLeftBiased` invariant (`psubtract` never returns `counter_ident`). A concrete `Finset` α instance is fully proven.

Condition-gated rewrite rules. The `LanguageDef pathMapLang` has ten rewrite rules, each with a list of `RelationQuery` premises ("`sub`", "`nsub`", "`disj`", "`ndisj`"). The full table for `pjoin` is:

Rule	Condition	Reduct
<code>PJoin_self</code>	$b \subseteq a$	a (Identity SELF)
<code>PJoin_other</code>	$a \subseteq b$	b (Identity COUNTER)
<code>PJoin_element</code>	$\neg(b \subseteq a) \wedge \neg(a \subseteq b)$	<code>PUnion(a, b)</code> (Element)

Analogous tables govern `pmeet` (three rules), `psubtract` (two rules, `None` when $a \subseteq b$), and `prestrict` (two rules, `None` when disjoint).

Complete NTT biconditionals. Let `relEnv` : `RelationEnv` encode which conditions hold for the current token pair, and write `cond_subba` for `applyPremisesWithEnv relEnv pathMapLa`. The main theorem for `pjoin` is:

Theorem 17 (`pathMap_pjoin_complete`, 0 sorries).

$$\llbracket \Diamond \varphi \rrbracket_{relEnv} (PJoin(a, b)) \iff (\text{cond_sub}_{ba} \wedge \varphi(a)) \vee (\text{cond_sub}_{ab} \wedge \varphi(b)) \vee (\text{cond_nsub_both} \wedge \varphi(PUnion(a, b)))$$

Each disjunct corresponds to exactly one `AlgebraicResult` case: Identity SELF, Identity COUNTER, and Element respectively. Analogous complete biconditionals are proven for `pmeet`, `psubtract`, and `prestrict` (`pathMap_pmeet_complete`, `pathMap_psubtract_complete`, `pathMap_prestrict_complete`).

The proofs use two infrastructure lemmas added to `MeTTaIL/Engine.lean`: `premiseStepWithEnv_relationQuery_mem` and `premiseStepWithEnv_congruence_mem`, which expose the (previously private) premise-step semantics as a public API. Binding preservation across premise chains is certified by `premisesEval_preserves_a/b`.

Key results summary:

- `pathMapGalois`: the auto-generated Galois connection $\Diamond \dashv \blacksquare$ for `pathMapLang` (0 sorries).
- `pathMap_pjoin_complete`: exact biconditional, all three `AlgebraicResult` outcomes, no case dropped.
- `pathMap_pmeet_complete`, `pathMap_psubtract_complete`, `pathMap_prestrict_complete`: same for the other three operations.

- **SubtractLeftBiased** instance: `psubtract` never returns `counter_ident`, proven for `Bool` and `Finset` α .

`PathMap` is the only OSLF instance in the formalization where the diamond theorem is complete in the sense that every reachable outcome is characterised: the biconditional leaves no case unaddressed.

11.7.1 Trie data structure (byte-indexed, coalgebraic)

The abstract `PathMapQuantale` typeclass is instantiated above for `Bool` and `Finset` α , but these are toy instances—the actual Hyperon runtime stores paths in byte-indexed tries. The formalization now includes a full trie stack (`PathMap/Trie/`, 970 lines, 0 sorries) providing the first concrete trie-backed `PathMapQuantale` instance.

Coalgebraic semantics. The canonical semantic object is a *coinductive trie* (`Trie/CoinductiveTrie.lean`, 226 lines):

```
def CTrie (V : Type*) := List UInt8 → Option V
```

A trie *is* its lookup function—the Brzozowski-derivative view [?]. Bisimilarity is extensional: $t_1 \sim t_2 \iff \forall p, t_1(p) = t_2(p)$. All algebraic laws reduce to pointwise `Option/Bool` identities by induction on the path p :

Law	Statement (up to Bisim)
<code>union_idem</code>	$t \cup t \sim t$
<code>union_assoc</code>	$(t_1 \cup t_2) \cup t_3 \sim t_1 \cup (t_2 \cup t_3)$
<code>inter_idem</code>	$t \cap t \sim t$
<code>inter_assoc</code>	$(t_1 \cap t_2) \cap t_3 \sim t_1 \cap (t_2 \cap t_3)$
<code>absorption</code>	$t \cup (t \cap s) \sim t$
<code>inter_union_distrib</code>	$t_1 \cap (t_2 \cup t_3) \sim (t_1 \cap t_2) \cup (t_1 \cap t_3)$

Brzozowski derivatives commute with all three operations (`deriv_union`, `deriv_inter`, `deriv_diff`).

Finite trie. The executable representation is an inductive `FTrie V` (`Trie/FiniteTrie.lean`, 224 lines) with sparse sorted child lists `List (UInt8 × FTrie V)`, matching the Rust `LineListNode` layout. Four operations—`join`, `meet`, `subtract`, `restrict`—merge child lists in lock-step on sorted keys.

Refinement homomorphism. The embedding $\iota : \text{FTrie} \hookrightarrow \text{CTrie}$ is simply $\iota(t) := t.\text{lookup}$ (`Trie/TrieRefinement.lean`, 243 lines). When both input tries are well-sorted (`FTrie.Sorted`), `join` commutes with the embedding:

Theorem 18 (`toCTrie_join`, 0 sorries). *For sorted t_1, t_2 , $\iota(t_1.\text{join } t_2) \sim \iota(t_1) \cup \iota(t_2)$.*

The proof proceeds by mutual induction on the trie and child-list structure, with a helper lemma `lookupChild_none_of_joinChildren` discharging the case where a key falls below all children in a merged list.

PathMapQuantale instance. `Trie/TriePathMapInstance.lean` (129 lines) wraps `FTrie` operations in `AlgebraicResult` with identity detection via structural `BEq`. Laws proven: `JoinIdem`, `MeetIdem`.

Commutativity caveat. `FTrie.join` is left-biased: when both tries have values at the same path, the left value is kept. `JoinComm` therefore holds only when `V` has at most one value (e.g. `V = Unit`).

Zipper iteration soundness. `Trie/TrieZipper.lean` (148 lines) provides `SimpleTrieZipper V`, a cursor over `FTrie.entries` that satisfies `ZipperIterationSound`: iterating via `toNextVal` visits exactly the stored values. This connects to the ZAM soundness theorems in `PathMap/ZipperExecution.lean`, which prove that any `ZipperIterationSound` implementation produces semantically identical results to a flat `RelationEnv`—and therefore the full OSLF type system (Galois connection, optimisation theorems) transfers automatically.

Summary. The trie stack adds 970 lines to the `PathMap` formalization (total now 4,086 lines, 0 sorries), providing the first `PathMapQuantale` instance backed by the actual trie data structure used in the Hyperon runtime rather than a flat-set mock.

12 Modal Hypercube and the Equational Center

The OSLF pipeline generates a type system from *any* `LanguageDef`—but *how many* type systems does a given language admit? Following Stay, Meredith & Wells [?], each rewrite rule generates a family of *modal type formers* parameterized by *sort assignments*: each rely-input variable and the output slot can independently be assigned `*` (type-level) or `□` (kind-level). Equational axioms force certain slots to agree, collapsing the naive 2^n -element cube to a smaller *equational center* $\mathcal{Z}$.

The file `Framework/ModalHypercube.lean` (0 sorries) formalizes Definition 5.1 and §5.6 of [?].

12.1 Core Definitions

Definition 19 (`HSort`). *The two-element sort universe $\mathbb{S} = \{*, \square\}$, formalized as an inductive type with `Fintype`, `DecidableEq`, and `BEq` instances.*

Definition 20 (`ModalPresentation n`). *A modal theory presentation with n sort slots and a decidable admissibility predicate `admissible : (Fin n → HSort) → Bool`.*

Definition 21 (`equationalCenter`). $\mathcal{Z} = \{\Sigma \in \mathbb{S}^n \mid \text{admissible}(\Sigma)\}$, computed as `Finset.univ.filter admissible`.

All proofs use `decide` (kernel-checked).

12.2 Worked Examples

Weak commutative monoid (§5.6.1 of [?]). Three slots (s_q, s_r, s_{out}), one commutativity equation $(qr) = (rq)$. Commutativity forces $s_q = s_r$; output agreement forces $s_{\text{out}} = s_q$. The naive 8-element cube collapses to 2.

Theorem 22 (`weakMonoid_center_card`, 0 sorries). $|\mathcal{Z}| = 2$ for the weak monoid presentation. Proven by *decide*.

ϱ -calculus communication modality (§5.12). Three slots ($s_n^{\text{Nm}}, s_p^{\text{Pr}}, s_{\text{out}}$), no equational constraints. Full 8-element cube survives; the (s_p, s_{out}) face recovers Barendregt’s Lambda Cube.

Theorem 23 (`rhoComm_center_card`, 0 sorries). $|\mathcal{Z}| = 8$ for the ϱ -calculus communication modality.

12.3 Application to the WM Calculus

The world-model calculus (`Framework/WMCalculusLanguageDef.lean`) has five core rewrite rules. Two are commutativity equations (`revision_comm`, `combine_comm`), each exhibiting the weak-monoid collapse pattern: three slots collapsing to two admissible assignments.

WM Rule	Slots	Naive	$ \mathcal{Z} $	Collapse
<code>evidence_add</code>	2	4	4	1×
<code>revision_comm</code>	3	8	2	4×
<code>combine_comm</code>	3	8	2	4×
<code>revision_assoc</code>	2	4	4	1×
<code>combine_zero</code>	1	2	2	1×
Product	11	2048	128	16×

Theorem 24 (`wmCore_collapse_ratio`, 0 sorries). The WM commutativity equations reduce the modal type-system family from 2048 to 128 vertices, a 16×

12.4 Application to PeTTa

The PeTTa artifact pipeline produces a concrete `LanguageDef` for each frozen PeTTa program via `toLanguageDef` (runtime) and `pettaSpaceToLangDef` (formal), connected by `Conformance/PeTTaArtifactBridge.lean`.

Both paths set `equations := []`, so every PeTTa modality gets its full naive cube—no equational collapse.

Theorem 25 (`unconstrained_center_eq_pow`, 0 sorries). For any unconstrained n -slot presentation, $|\mathcal{Z}| = 2^n$.

This result identifies a *semantic enrichment opportunity*: bidirectional PeTTa rewrite rules (used in both directions) are effectively equations. Detecting these

in the lowering path and adding them to the `equations` field would enable equational collapse, reducing the type-system family size for PeTTa programs with symmetric rules.

12.5 Modal Profile Artifact

`ModalProfile.ofLanguageDef` computes a stored modal profile from any `LanguageDef`, recording per-rule slot counts, the naive product, and whether equations exist. This artifact lives alongside the execution contract and scope contract in the PeTTa artifact bundle, making the type-system family structure inspectable without re-running the analysis.

Summary. The modal hypercube formalization adds 24 kernel-checked theorems (0 sorries) in a single file, applying the equational center construction to the WM calculus (16× collapse from commutativity), PeTTa (no collapse, enrichment opportunity flagged), and the ϱ -calculus (full cube, Barendregt face identified).

13 Verified Formula Checker

The file `Formula.lean` provides a verified bounded model checker for OSLF formulas (1,204 lines, 0 sorries). The formula type `OSLFFormula` supports $\Diamond$, $\blacksquare$, $\wedge$, $\vee$, $\implies$, $\top$, $\perp$, and atomic predicates.

Theorem 26 (Checker Soundness, 0 sorries). *If the checker returns `.sat` for formula φ at term p , then $p \models \varphi$ in the denotational semantics.*

The checker supports:

- `checkWithPred`: checking with external predecessor functions, enabling bounded $\blacksquare$ verification;
- `aggregateBox`: universal checking of $\blacksquare\varphi$ over a predecessor list, with proven soundness (`aggregateBox_sat`).

14 Categorical Lift and Presheaf Topos

The file `Framework/CategoryBridge.lean` (1,603 lines, 0 sorries) lifts the Set-level OSLF construction into Mathlib’s categorical infrastructure, culminating in the presheaf topos $P(T) = \mathbf{Set}^{T^{op}}$ over the constructor category.

14.1 Modal Adjunction

The predicate type `Pattern  $\rightarrow$  Prop` carries conflicting category instances (`Preorder.smallCategory` vs. `CategoryTheory.Pi`). We introduce a type wrapper `PredLattice` to disambiguate, then lift:

Listing 11: Modal adjunction (CategoryBridge.lean)

```
noncomputable def langModalAdjunction (lang : LanguageDef) :  
  (langGaloisL lang).monotone_l.functor |-  
  (langGaloisL lang).monotone_u.functor :=  
  (langGaloisL lang).adjunction
```

This provides a categorical **Adjunction** between the $\diamond$ and $\blacksquare$ endofunctors on the predicate preorder category, for any **LanguageDef**.

14.2 Presheaf Topos and Yoneda Representables

Following Williams and Stay [?], we construct the presheaf topos $P(T)$ as a lambda-theory with equality:

Listing 12: Presheaf topos (CategoryBridge.lean)

```
def languagePresheafLambdaTheory (lang : LanguageDef) :  
  LambdaTheoryWithEquality where  
  Obj := Functor (Opposite (ConstructorObj lang)) Type  
  -- CCC structure via Mathlib
```

The Yoneda embedding sends each sort s to a representable presheaf $y(s) \in P(T)$. The subobject classifier Ω is constructive via sieves, and the characteristic-map equivalence $\text{Sub}(y(s)) \cong (y(s) \rightarrow \Omega)$ is proven (`languageSortFiber_characteristicEquiv`).

A bridge maps Set-level predicates ($\text{Pattern} \rightarrow \text{Prop}$) into presheaf-level subobjects $\text{Sub}(y(s))$ under a naturality condition (`languageSortFiber_ofPatternPred`). The reduction relation is internalized as a **Subfunctor** of the pair presheaf with graph-object diamond/box characterization (`langDiamondUsing_iff_exists_graphStep`, `langBoxUsing_iff_forall_graphIncoming`).

14.3 Current Scope and the Full Grothendieck Construction

Our presheaf topos construction provides the categorical *home* for OSLF types but does not yet fully internalize the modalities. Specifically:

What is proven. $P(T)$ exists as a CCC with finite limits, subobject classifier, and exponentials (all via Mathlib). Yoneda representables, fiber characterization, and the predicate-to-subfunctor bridge are proven. The reduction relation is internal to $P(T)$ as a subfunctor. Beck–Chevalley instances for specific squares (COMM, substitution) are proven at the subobject level.

What remains. The $\diamond/\blacksquare$ operators are defined as Set-level endomorphisms on $(\text{Pattern} \rightarrow \text{Prop})$ and have not yet been lifted to internal natural transformations $\Omega \rightarrow \Omega$ in $P(T)$. This would require:

1. *Internalizing the reduction span:* replacing the constant presheaf with a genuine span $y(\text{Proc}) \leftarrow R \rightarrow y(\text{Proc})$ in $P(T)$, where the edge set varies with the generalized-element context.

2. *Internal existence/universal image*: defining $\Diamond/\blacksquare$ as $\exists_{\text{src}} \circ \text{tgt}^*$ and $\forall_{\text{tgt}} \circ \text{src}^*$ at the subfunctor level.
3. *Agreement*: proving that the internal modalities, evaluated at representable sorts, reproduce the Set-level modalities.

The full Grothendieck construction for Native Type Theory [?] is now formalized: the strict NTT claim tracker (`Framework/NTTClaimTracker.lean`) shows 12/12 claims resolved (all proven, including a necessity witness for the `types.Nonempty` guard), including Prop 12 (indexed adjoints + Beck–Chevalley), Prop 14 (cosmic fibration), Prop 17 (reification right adjoint), Def 21 (codomain fibration with Cartesian lifts via pullbacks), Sec 4 (image-comprehension with full $\leftrightarrow$ characterization), and Thm 23 (internal language package with functorial laws for identity and composition of theory morphisms). The sole assumption-scoped item is Π/Σ type formation under a `types.Nonempty` guard, whose necessity is proven by `types_nonempty_necessary_for_piSigma`. Kripke–Joyal forcing semantics, not currently in Mathlib, would be needed to fully connect the internal language of $P(T)$ to OSLF typing judgments.

15 Paper-Parity Endpoints

We provide a *paper-parity canonical package* that bundles the core claims of the three OSLF source papers into a single Lean record type, instantiable on any `LanguageDef`.

Listing 13: Unified paper-parity package (`CoreMain.lean`)

```
-- The unified endpoint for review:
#check @Mettapedia.OSLF.coreMain_paper_parity_full_package
-- Type: (lang : LanguageDef) -> AtomSem
--       -> ScopedConstructorPred lang -> ...
--       -> CoreMainPaperParityCanonicalPackage lang ...
```

The package contains four milestone bundles:

- M1 Full presheaf Grothendieck category** for any language (`fullPresheafGrothendieckCategory`).
- M2 Scoped/full comparison**: roundtrip between `ScopedConstructorPred` and `FullPresheafGrothendieckObj` (`scoped_full_scoped_obj_roundtrip`).
- M3 Internal language**: frame-derived $\rightarrow/\neg$, $\top/\perp/\wedge/\vee$, and assumption-guarded Π/Σ via `topos_full_internal_logic_bridge_package`.
- M4 TOGL graph-modal bridge**: n -step graph paths $\leftrightarrow$ n -fold relational composition (`graphChainN_iff_relCompN`), modal iteration $\Diamond^n \leftrightarrow$ graph chains (`diamondIterN_iff_graphChainN`), and internal subfunctor correspondence (`togl_internal_graph_correspondence_layer`).

15.1 Traceability: Paper Claims to Lean Theorems

A machine-checked paper-claim tracker (`Framework/PaperClaimTracker.lean`) maps 27 verifiable claims from three source papers to Lean theorem names (plus a strict NTT tracker with 12 claims in `Framework/NTTClaimTracker.lean`):

Paper	Loc	Claim	Status
oslf.pdf	Def 1	Rewrite system	proven
oslf.pdf	§3	Step-future $\Diamond$	proven
oslf.pdf	§3	Step-past $\blacksquare$	proven
oslf.pdf	§4	Galois $\Diamond \dashv \blacksquare$	proven
oslf.pdf	§6	OSLF type system output	proven
oslf.pdf	Thm 1	Behavioral $\Rightarrow$ same types (fwd)	proven
oslf.pdf	Thm 1	Full substitutability (image-finite)	assumption-scoped [†]
oslf.pdf	§11	ϱ -calculus instance	proven
oslf.pdf	§11	λ -calculus instance	proven
oslf.pdf	§11	Petri net instance	proven
oslf.pdf	§12.1	Compile-time firewall	proven
oslf.pdf	§12.2	Race detection	proven
oslf.pdf	§12.3	Secrecy	proven
oslf.pdf	§13.1	Dependent/parametric extension	proven
NTT.pdf	§3	Native type = (sort, predicate) pair	proven
NTT.pdf	Def 11	Predicate fibration π_Ω	proven
NTT.pdf	Prop 12	Indexed adjoints + Beck–Chevalley	proven
NTT.pdf	Prop 14	Cosmic fibration (Frame fibers)	proven
NTT.pdf	Prop 17	Reification right adjoint	proven
NTT.pdf	Def 21	Codomain fibration + Cartesian lifts	proven
NTT.pdf	§4	Image-comprehension adjunction (full iff)	proven
NTT.pdf	Thm 23	Internal language package + functorial laws	proven
NTT.pdf	§4	Internal language: $\top/\perp/\wedge/\vee$	proven
NTT.pdf	§4	Frame-derived $\rightarrow/\neg$	proven
NTT.pdf	§4	Π/Σ type formation	assumption-scoped
NTT.pdf	§5	Theory morphism Π/Ω	proven
NTT.pdf	§5	Colax Π /Prop rules	proven
NTT.pdf	§5	Necessity: nonempty-family guard	proven
togl.pdf	§2	Graph reduction structure	proven
togl.pdf	§3	Graph-modal bridge	proven
togl.pdf	§4	N -step paths = relational composition	proven
togl.pdf	§4	Modal iteration $\Diamond^n$ = graph chains	proven
togl.pdf	§5	Internal subfunctor correspondence	proven

[†]A decidable sufficient condition (`LangPredFiniteSafe`, Theorem 15) discharges the predecessor-finiteness assumption for all languages satisfying the structural criterion; the ϱ -calculus is provably excluded.

26 claims are fully proven; 1 is assumption-scoped with proven necessity counterexamples showing the assumptions cannot be globally discharged, plus a

positive sufficient condition (Section 10.2) that discharges predecessor-finiteness for structurally well-behaved languages:

- `types_nonempty_necessary_for_piSigma`: for empty families, $\bigcap \emptyset = \top$ and $\bigcup \emptyset = \perp$, so $\top \leq \perp$ fails in any nontrivial Frame.
- `hClosed_necessary_for_fragment`: a seed-restricted fragment is not closed under `ScopedReachable`.
- `not_global_hImageFinite_rhoCoreStarRel`: the core-star endpoint relation is not globally image-finite (via `addRightZeroNest` injection).
- `not_global_hPredFinite_langReduces_rhoCalc`: the canonical ϱ -calculus COMM rule erases the channel name, yielding infinitely many distinct one-step predecessors for a fixed target.
- `rhoCalc_not_langPredFiniteSafe`: the COMM rule violates the structural sufficient condition (RHS contains `.subst/.collection` nodes; variable n erased), confirming that the negative and positive results meet at the ϱ -calculus boundary.

Robustness canaries (`Framework/PaperParityCanaries.lean`) instantiate the full package on ϱ -calculus and λ -calculus (positive), plus a negative canary demonstrating that the `hClosed` assumption does real work.

16 Status and Roadmap

16.1 Milestones

- ✓ **Milestone 1:** Executable ϱ -calculus engine. `reduceStep` computes all one-step reducts; `reduceStep_sound` proven with 0 sorries; 6 executable tests pass.
- ✓ **Milestone 2:** Generic MeTTaIL framework. Language-parametric pattern matching (`Match.lean`) and rewriting (`MeTTaIL/Engine.lean`) for any `LanguageDef`. 8-test agreement suite confirms equivalence with specialized engine.
- ✓ **Milestone 3:** OSLF type synthesis. `langOSLF` mechanically generates an `OSLFTypeSystem` from any `LanguageDef` with auto-proven Galois connection. `GenHasType` provides generated typing rules. Three-layer bridge established.
- ✓ **Milestone 4:** Correctness infrastructure. Totalized `Match.lean` (no `partial def` in core matching). Least authored `ContextualStep.Step` with exact executable compilation. Enhanced formula checker with predecessor-based ■ checking.

- ✓ **Milestone 5:** Language instances. Seven languages—lambda calculus, Petri nets, ϱ -calculus, TinyML, MeTTaMinimal, MeTTaFull, and PathMap—each with auto-generated `OSLFTypeSystem` and Galois connection.
- ✓ **Milestone 6:** Presheaf-primary categorical lift. Interface-selected bases (`SortCategoryInterface`) and presheaf-primary predicate fibrations are wired via `CategoryBridge`, including representable-fiber characteristic equivalences and checker-to-fiber soundness.
- ✓ **Milestone 7:** Graph-object BC path and end-to-end clients. `ConstructorCategory`: free category on sort-crossing constructors. `ConstructorFibration`: `SubobjectFibration` + `ChangeOfBase` with proven adjunctions. `ModalEquivalence` and `DerivedTyping`: modal/typing synthesis from constructor structure. `ToposReduction/BeckChevalleyOSLF`: reduction graph objects, explicit substitution squares, and graph-level $\diamond/\blacksquare$ compatibility consumed by TinyML, MeTTaMinimal, and MeTTaFull checker-to-semantics theorems. All 0 sorries in this core track.
- ✓ **Milestone 8:** Paper-parity canonical package. `coreMain_paper_parity_full_package` bundles M1 (Grothendieck category), M2 (scoped/full comparison), M3 (internal language), and M4 (TOGL graph-modal bridge) into one endpoint for any `LanguageDef`. 27 paper claims tracked; 25 fully proven, 2 assumption-scoped with proven necessity counterexamples. Robustness canaries on ϱ -calculus and λ -calculus (Section 15).
- ✓ **Milestone 9:** Strict NTT parity and strengthened endpoints. All 12 strict NTT claims resolved (all proven, including a necessity witness for the nonempty-family guard). Three endpoints strengthened beyond paper parity: (a) Thm 23 internal language package with functorial laws (identity/composition preservation of $\Pi/\Omega/\text{Prop}$); (b) Def 21 Cartesian lifts via pullbacks with universal factorization; (c) image-comprehension $\text{full} \leftrightarrow (\text{hom-set characterization of } i \dashv c)$.

17 PLN as a Proof System

The WM-PLN evidence algebra is formalized as a *classical proof system* under the Curry-Howard correspondence, with KS axiom foundations, hypercube classification, GSLT weight-map bridge, and certifiable runtime implementations. The showcase file `Logic/PLNProofSystemShowcase.lean` (0 sorries) serves as the single entry point, importing 10 core files and re-exporting 15 key theorems with narrative documentation.

17.1 Curry-Howard Correspondence for Evidence

`Logic/EvidenceProofSystem.lean` (0 sorries) establishes that the WM evidence algebra satisfies all four structural rules of classical sequent calculus:

Proof Rule	Evidence Operation	Theorem
Cut	Revision (addition)	<code>cut_rule</code>
Weakening	Forgetting	<code>weakening_preserves_validity</code>
Exchange	Commutativity	<code>exchange_rule</code>
Contraction	Self-revision	<code>contraction_doubles_ess</code>

Theorem 27 (`structural_rules_summary`, 0 sorries). *The evidence algebra satisfies cut, weakening, exchange, and contraction over $\mathbf{BinEvNat}$.*

17.2 KS-Evidence-Measure Triangle

`Logic/KSEvidenceMeasureBridge.lean` (0 sorries) connects the evidence algebra to the Knuth-Skilling axiomatization and measure theory:

1. $\mathbf{BinEvNat}$ with `hplus` is an ordered commutative monoid satisfying the KS axioms (monotone, associative, separating).
2. The representation $\Theta(e) = e.pos + e.neg$ is additive (`$\Theta\_additive$`) and normalized (`$\Theta\_zero$`).
3. The measure bridge factors through Θ via Dirac weighting: $\mu_W = \sum_q \Theta(\text{extract } W \ q) \cdot \delta_q$.
4. σ -additivity requires countable additivity—the finitely additive case does NOT extend (`$\sigma\_additivity\_boundary$` , imported from the KS counterexample formalization).

17.3 Probability Hypercube Classification

`Logic/WMHypercubeClassification.lean` and `Logic/ModalProbabilityBridge.lean` (0 sorries each) classify 4 WM regimes at 4 probability vertices:

WM Regime	Logic	Representation
Additive (classical)	Boolean	Point
Overlap-aware	Boolean	Bounds
Evidence-tracking (PLN)	Heyting	Bounds
Trust-gated	Heyting	Point

Theorem 28 (`full_classification_injective`, 0 sorries). *The map from 4 WM regimes to 13-axis probability vertices is injective.*

The $\mathbb{C}$ carrier (`Logic/ComplexEvidenceCarrier.lean`, 0 sorries) sits at the quantum vertex: `orthomodular` distributivity, no lattice structure. The `AdditiveWorldModel` works unchanged over $\mathbb{C}$; probability is recovered via the Born rule $|\cdot|^2$.

17.4 GSLT Weight Map Bridge

`Logic/GSLTWeightMapBridge.lean` (0 sorries) shows that the GSLT weight map on rewrite states IS the WM extract function: `EvidenceAssignment.weight = AdditiveWorldModel.extract`, with `weight_add = extract_add`.

17.5 Calculus Soundness

`PLN/WorldModel/WMCalculusSoundness.lean` (0 sorries) defines the soundness criterion for the WM calculus:

1. Define `WMCalculus State Query Ev` (the operational rules: revise, extract, forget).
2. Prove `CalculusSound` (revision = addition, extraction = algebraic extract).
3. All algebraic properties transfer automatically: `extract_add`, sequential composition, zero preservation.

Theorem 29 (`soundness_guarantees`, 0 sorries). *A sound calculus inherits extraction correctness, revision correctness, and the `extract_add` law from the WM algebra.*

17.6 Universality and Empirical Validation

`PLN/WorldModel/UniversalEnsembleWM.lean` (0 sorries) proves that every typed observation space generates a unique additive world model (existence + uniqueness). `PLN/Evidence/SourceReliability.lean` (0 sorries) adds a Dawid-Skene reliability layer with per-source accuracy profiles. `Logic/PLNClassicExamples.lean` (0 sorries) validates deduction-revision, raven induction, and frame/relevance examples with kernel-checked stamp disjointness.

18 Conclusion

We have presented the first machine-checked formalization of the OSLF algorithm, spanning 44,000+ lines of Lean 4 across 124 files with 0 sorries and 0 custom axioms. The formalization demonstrates that the entire pipeline from rewrite rules to spatial-behavioral types can be made rigorous in a modern proof assistant.

The Galois connection at the heart of OSLF is proven at three levels: (1) directly for the ϱ -calculus, (2) categorically for any reduction span via adjoint composition, and (3) lifted to a Mathlib `Adjunction` between endofunctors on the predicate preorder category. All three levels are shown to agree.

The constructor category (Section 4) provides a genuine categorical backbone: sort-crossing constructors become morphisms, the fibered change-of-base

gives the adjoint triple $\exists_f \dashv f^* \dashv \forall_f$ for each constructor, and the modal operators $\Diamond/\blacksquare$ are shown to be the typing actions of quoting/reflecting constructors. The derived typing rules (Section 4.4) make explicit that the assignment $\text{NQuote} \mapsto \Diamond$, $\text{PDrop} \mapsto \blacksquare$ is not ad hoc but follows from the constructor’s position in the category. The Beck–Chevalley analysis (Section 4.5) shows that while substitution commutes with change-of-base in the COMM-specific case (proven via `comm_preserves_type`), the universal condition fails—a mathematically interesting negative result proven by explicit counterexample.

The presheaf topos $P(T) = \mathbf{Set}^{T^{\text{top}}}$ over the constructor category is constructed with Yoneda representables, subobject classifier, and fiber characterization (Section 14). The full presheaf Grothendieck category for Native Type Theory [?] is now formalized with proven associativity and identity laws, scoped/full comparison roundtrip, and internal-language bridge (Section 15). All 12 strict NTT claims are resolved: the codomain fibration has Cartesian lifts via pullbacks, the image-comprehension adjunction has a full hom-set $\leftrightarrow$ characterization, and the internal language package satisfies identity and composition laws. The remaining frontier item—lifting Set-level modalities into genuine internal $\Omega \rightarrow \Omega$ natural transformations in $P(T)$ —is precisely characterized in Section 14.3.

The full type synthesis pipeline is validated by seven language instances (ϱ -calculus, lambda calculus, Petri nets, TinyML, MeTTaMinimal, MeTTaFull, and PathMap), each with auto-proven Galois connections. TinyML demonstrates the framework’s ability to handle multi-sort CBV languages with binders, conditionals, and sort-crossing constructors, mirroring the ϱ -calculus’s NQuote/PDrop structure with Thunk/Inject. The MeTTa instances close the loop from the interpreter specification back to the categorical semantics. PathMap extends the framework to *condition-gated* rewriting with a ternary result type, and is the first instance where the NTT diamond is proven as a complete biconditional across all `AlgebraicResult` outcomes (Section 11.7).

A declarative reduction relation provides an engine-independent specification, and the executable engines are proven both sound and complete with respect to it. All nine *core* implementation milestones are complete and sorry-free. The paper-parity canonical package (Section 15) bundles 27 verifiable claims from three source papers—25 fully proven, 2 assumption-scoped with proven necessity counterexamples—into a single machine-checked endpoint. The strict NTT tracker independently confirms 12/12 claims resolved at theorem level. The strict paper-claim tracker (`Framework/PaperClaimTracker.lean`) and robustness canaries (`Framework/PaperParityCanaries.lean`) provide ongoing regression coverage for the formalization’s alignment with the mathematical literature.